Assignment 03

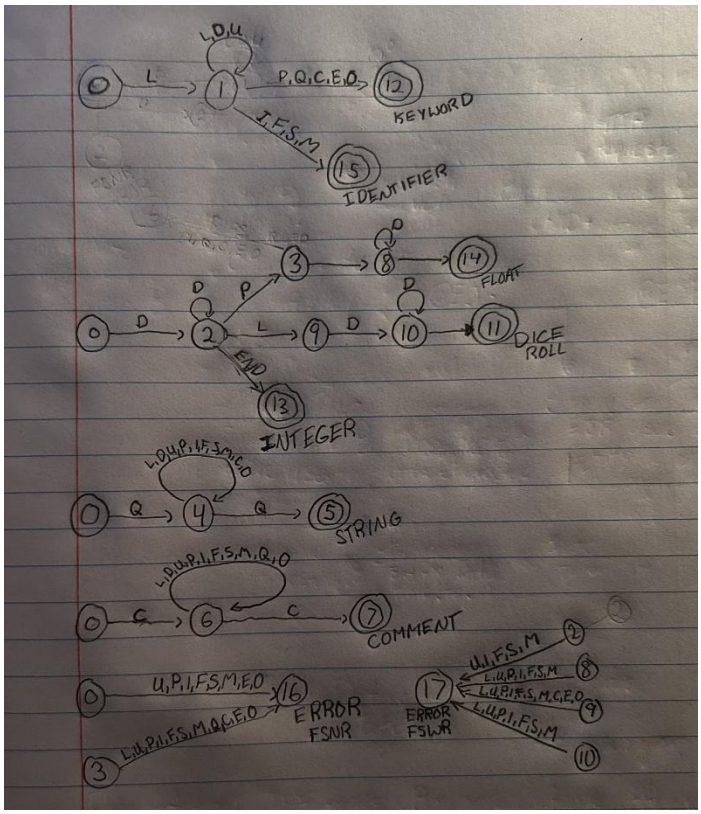
Regular Expressions
Transitional Diagrams
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Char Classes

L = Letter
D = Digit
U = Underscore (_)
P = Period (.)
I = Dollar Sign ($)
F = Modulus (%)
S = At Sign (@)
M = And Sign (&)
Q = Quotation (“)
C = Hashtag (#)
E = End
O = Other

Accepting States

5 – STRING (FSNR)
7 – COMMENT (FSNR)
11 – DICE ROLL (FSWR)
12 – KEYWORD (FSWR)
13 – INTEGER (FSWR)
14 – FLOAT (FSWR)
15 – IDENTIFIER (FSNR)



[DFA]

# STATE	L	D	U	P	I	F	S	M	Q	C	E	O	STATE_TYPE	ACTION
0	1	2	16	16	16	16	16	16	4	6	16	16	NOFS	NULL
1	1	1	1	12	15	15	15	15	12	12	12	12	NOFS	NULL
2	9	2	17	3	17	17	17	17	13	13	13	13	NOFS	NULL
3	16	8	16	16	16	16	16	16	16	16	16	16	NOFS	NULL
4	4	4	4	4	4	4	4	4	5	4	16	4	NOFS	NULL
5	18	18	18	18	18	18	18	18	18	18	18	18	FSNR	STRING
6	6	6	6	6	6	6	6	6	6	7	16	6	NOFS	NULL
7	18	18	18	18	18	18	18	18	18	18	18	18	FSNR	COMMENT
8	17	8	17	17	17	17	17	17	14	14	14	14	NOFS	NULL
9	17	10	17	17	17	17	17	17	17	17	17	17	NOFS	NULL
10	17	10	17	17	17	17	17	17	11	11	11	11	NOFS	NULL
11	18	18	18	18	18	18	18	18	18	18	18	18	FSWR	DICE
12	18	18	18	18	18	18	18	18	18	18	18	18	FSWR	KEYWORD
13	18	18	18	18	18	18	18	18	18	18	18	18	FSWR	INTEGER
14	18	18	18	18	18	18	18	18	18	18	18	18	FSWR	FLOAT
15	18	18	18	18	18	18	18	18	18	18	18	18	FSNR	IDENTIFIER
16	18	18	18	18	18	18	18	18	18	18	18	18	FSNR	ERROR
17	18	18	18	18	18	18	18	18	18	18	18	18	FSWR	ERROR

Tokens	Regular Expressions	Example
String Literal	“[^”]*”	“Hello World!”
Comment	#[^#]*#	# ... #
Dice Literal	[0-9]+d[0-9]+	2d6
Keyword	[A-Za-z][A-Za-z0-9_]* + Lookup	STEL_int, STEL_float, ...
Integer Literal	[0-9]+	123
Float Literal	[0-9]+\.[0-9]+	12.32
Integer Identifier	[A-Za-z][A-Za-z0-9_]*$	num$
Float Identifier	[A-Za-z][A-Za-z0-9_]*%	floatNum%
String Identifier	[A-Za-z][A-Za-z0-9_]*@	name@
Method Identifier	[A-Za-z][A-Za-z0-9_]*&	main&